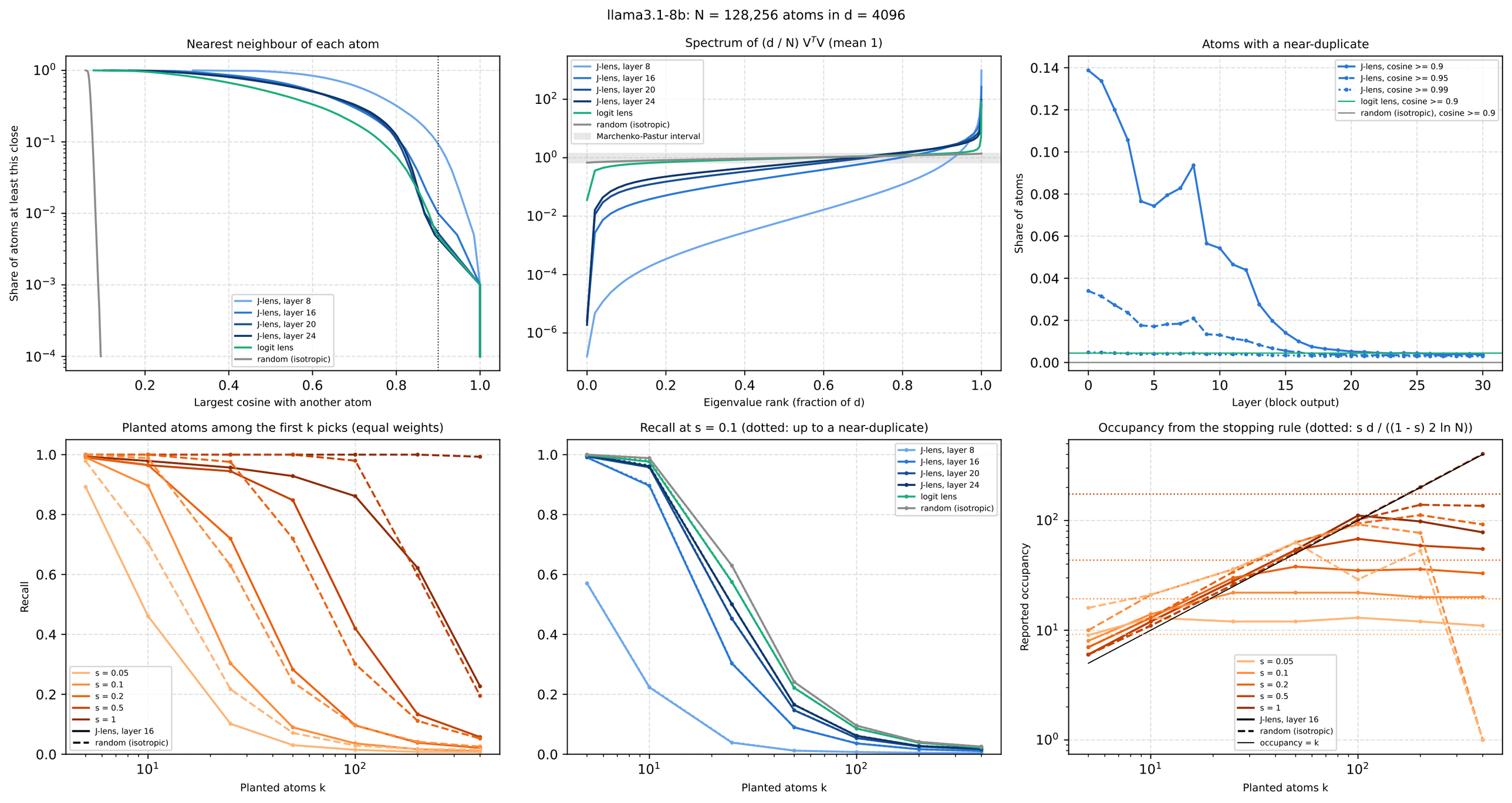
llama3.1-8b: N = 128,256 atoms in d = 4096
Nearest neighbour of each atom
Share of atoms at least this close
Largest cosine with another atom
J-lens, layer 8
J-lens, layer 16
J-lens, layer 20
J-lens, layer 24
logit lens
random (isotropic)
Spectrum of (d / N) V^T V (mean 1)
Marchenko-Pastur interval
Eigenvalue rank (fraction of d)
Atoms with a near-duplicate
Share of atoms
Layer (block output)
J-lens, cosine >= 0.9
J-lens, cosine >= 0.95
J-lens, cosine >= 0.99
logit lens, cosine >= 0.9
random (isotropic), cosine >= 0.9
Planted atoms among the first k picks (equal weights)
Recall
Planted atoms k
s = 0.05
s = 0.1
s = 0.2
s = 0.5
s = 1
Recall at s = 0.1 (dotted: up to a near-duplicate)
Occupancy from the stopping rule (dotted: s d / ((1 - s) 2 ln N))
Reported occupancy
occupancy = k